

	A	B	C	D	E	F	G	H
1	Power Budget Example							
2	Team Number:	Team 209						
3	Project Name:	Smart Door Sensor						
4	Team Member Names:	Andrew, Bryce, Matthew, Dylan						
5	Version:	v1.0						
6								
7	A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
8	All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
9		Op Amp	MCP6004-I/P	+1.6 - 6V	1	0.17	0.17	mA
10		Relay Switch	CST10T2CR	+0.1 - 20V	1	20	20	mA
11		4166 Part Passive Buzzer 5V-I	4166-PARTPASSI	+2.0 - 6.0V	1	30	30	mA
12		PIC18F57Q43 Curiosity Nano	PIC18F57Q43	+1.8 - 6.0V	1	0.1	0.1	mA
13								
14	B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
15	+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
16		PSoC™ 4 BLE Module	PIC18F57Q10PIC18F	+1.8 - 6.0V	1	200	200	mA
17		Op Amp	MCP6004-I/P	+1.6 - 6V	1	0.17	0.17	mA
18		Relay Switch	CST10T2CR	+0.1 - 20V	1	20	20	mA
19		4166 Part Passive Buzzer 5V-I	4166-PARTPASSI	+2.0 - 6.0V	1	30	30	mA
20						Subtotal	250.17	mA
21						Safety Margin	25%	
22						Total Current Required on +5V Rail	312.7125	mA
23								
24	c2. Regulator or Source Choice	5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
25						Total Remaining Current Available on +5V Rail	1187.2875	mA
26	C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each							
27	Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
28	+5V Power Rail	5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
29								
30	D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need							
31	External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
32	Power Source 1 Selection	Plug-in Wall Supply	L6R36-090	264VAC	+24V	3000	3000	mA
33								
34								
35	Power Rails Connected to External Power Source 1	5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
36								
37						Total Remaining Current Available on External Power Source 1	1500	mA
38								
39	External Power Source 2	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
40	Power Source 2 Selection	Battery	AA	+120V	1.5V	25000	25000	mA
41								
42	Power Rails Connected to External Power Source 2	5V regulator	AP7375-50SA-7	+5V - 45V	1	1500	1500	mA
43								
44								
45						Total Remaining Current Available on External Power Source 2	23500	mA
46								

	A	B	C	D	E	F	G	H
47	E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the battery by indicating the capacity in mAh.							
48	Component Name		Part Number	SupplyVoltageRange	Capacity(mAh)		Required	
49	Notes							
50	External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).							
51	If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit							